

OptiRBC

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theory <i>AbstractDomainModel</i>		
imports <i>Complex-Main</i>		
begin		

This is the abstract domain model used in the Ivy specification

```
locale abstract-domain-model =  
  fixes broadcaster :: 'p::finite  
    and faulty :: 'p set  
    and opt-quorum-member :: 'opQ  $\Rightarrow$  'p  $\Rightarrow$  bool  
    and maj-quorum-member :: 'vtQ  $\Rightarrow$  'p  $\Rightarrow$  bool  
    and classic-quorum-member :: 'quQ  $\Rightarrow$  'p  $\Rightarrow$  bool  
    and amplification-quorum-member :: 'amQ  $\Rightarrow$  'p  $\Rightarrow$  bool  
    and commit-quorum-member :: 'coQ  $\Rightarrow$  'p  $\Rightarrow$  bool  
assumes broadcaster-not-in-classic-quorum:  
   $\neg$  classic-quorum-member quQ broadcaster  
  and broadcaster-not-in-maj-quorum:  
   $\neg$  maj-quorum-member vtQ broadcaster  
  and broadcaster-not-in-opt-quorum:  
   $\neg$  opt-quorum-member opQ broadcaster  
  and non-faulty-commit-quorum:  
   $\exists$  coQ.  $\forall$  p. commit-quorum-member coQ p  $\longrightarrow$  p  $\notin$  faulty  
  and non-faulty-amplification-quorum:  
   $\exists$  amQ.  $\forall$  p. amplification-quorum-member amQ p  $\longrightarrow$  p  $\notin$  faulty  
  and non-faulty-maj-quorum:  
   $\exists$  vtQ.  $\forall$  p. maj-quorum-member vtQ p  $\longrightarrow$  p  $\notin$  faulty  
  and non-faulty-classic-quorum:  
   $\exists$  quQ.  $\forall$  p. classic-quorum-member quQ p  $\longrightarrow$  p  $\notin$  faulty  
  and opt-quorum-has-nonfaulty-member:  
   $\exists$  p. p  $\notin$  faulty  $\wedge$  opt-quorum-member opQ p  
  and opt-quorum-intersection:
```

```

    broadcaster ∈ faulty → (∃ p. p ∉ faulty ∧ opt-quorum-member opQ1 p ∧
    opt-quorum-member opQ2 p)
  and opt-maj-quorum-intersection:
    broadcaster ∈ faulty → (∃ p. p ∉ faulty ∧ opt-quorum-member opQ p ∧
    maj-quorum-member vtQ p)
  and opt-quorum-intersects-classic-quorum:
    broadcaster ∈ faulty → (∃ p. p ∉ faulty ∧ opt-quorum-member opQ p ∧ clas-
    sic-quorum-member quQ p)
  and classic-quorum-intersection:
    broadcaster ∈ faulty → (∃ p. p ∉ faulty ∧ classic-quorum-member quQ1 p ∧
    classic-quorum-member quQ2 p)
  and quorum-not-faulty:
    ∃ p. p ∉ faulty ∧ classic-quorum-member Q p
  and maj-quorum-has-nonfaulty-member:
    ∃ p. p ∉ faulty ∧ maj-quorum-member vtQ p
  and amplification-quorum-has-nonfaulty-member:
    ∃ p. p ∉ faulty ∧ amplification-quorum-member amQ p
  and opt-quorum-contains-maj-quorum:
    broadcaster ∈ faulty → (∃ vtQ. ∀ p. maj-quorum-member vtQ p → p ∉ faulty
    ∧ opt-quorum-member opQ p)
  and commit-quorum-contains-non-faulty-amplification-quorum:
    ∃ amQ. ∀ p. amplification-quorum-member amQ p → p ∉ faulty ∧ com-
    mit-quorum-member coQ p

end
theory AxiomaticDomainModel
  imports AbstractDomainModel
begin

```

In this theory we define types of quorums corresponding to the various thresholds used by the optimistic RBC algorithm. This will then allow us to do proof abstractly, just relying on intersection properties and not on cardinalities.

1 The domain model, axiomatically

```

typedecl party

```

```

axiomatization faulty :: party set and broadcaster :: party
  where party-finite: finite (UNIV :: party set)

```

```

instantiation party :: finite
begin

```

```

instance
  by intro-classes (simp add: party-finite)

```

```

end

```

definition $n :: \text{nat}$ **where**
 $n \equiv \text{card } (\text{UNIV} :: \text{party set})$

definition $f :: \text{nat}$ **where**
 $f \equiv \text{card } \text{faulty}$

axiomatization **where**
fault-bound: $3 * f < n$
and *two-parties*: $2 \leq n$

typedef *opt-quorum* =
 $\{S :: \text{party set. } \text{broadcaster} \notin S \wedge \text{card } S \geq \max 1 \lceil \text{real } (n + 2 * f - 2) / 2 \rceil\}$
proof –

Proof sketch: $\text{UNIV} - \{\text{broadcaster}\}$ excludes *broadcaster* and has cardinality $n - 1$. The global inequalities *fault-bound* and *two-parties* make this large enough for the optimistic threshold and ensure that it is nonempty.

let $?S = \text{UNIV} - \{\text{broadcaster}\} :: \text{party set}$
have *card-S*: $\text{card } ?S = n - 1$
by (*simp add: n-def*)
have *ge-one*: $\text{card } ?S \geq 1$
proof –
have $n - 1 \geq 1$
using *two-parties* **by** *linarith*
then show *?thesis*
by (*simp add: card-S*)
qed
have *thresh*: $\text{card } ?S \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
proof –
have $\text{real } (n - 1) \geq \text{real } (n + 2 * f - 2) / 2$
using *fault-bound* **by** *linarith*
then have $\lceil \text{real } (n - 1) \rceil \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
by (*intro ceiling-mono*)
then have $n - 1 \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
by *simp*
then show *?thesis*
by (*simp add: card-S*)
qed
show $\exists S :: \text{party set.}$
 $S \in \{S :: \text{party set. } \text{broadcaster} \notin S \wedge \text{card } S \geq \max 1 \lceil \text{real } (n + 2 * f - 2) / 2 \rceil\}$
using *ge-one thresh* **by** (*intro exI[of - ?S]*) *simp*
qed

typedef *maj-quorum* =
 $\{S :: \text{party set. } \text{broadcaster} \notin S \wedge \text{card } S \geq \lceil \text{real } n / 2 \rceil\}$
proof –

Proof sketch: $UNIV - \{broadcaster\}$ again provides all non-broadcaster parties. The global inequality $2 \leq n$ ensures that this set reaches the vote threshold.

```

let ?S = UNIV - {broadcaster} :: party set
have card-S: card ?S = n - 1
  by (simp add: n-def)
have thresh: card ?S >= ⌈real n / 2⌉
proof -
  have real (n - 1) >= real n / 2
    using two-parties by linarith
  then have ⌈real (n - 1)⌉ >= ⌈real n / 2⌉
    by (intro ceiling-mono)
  then have n - 1 >= ⌈real n / 2⌉
    by simp
  then show ?thesis
    by (simp add: card-S)
qed
show ∃ S :: party set.
  S ∈ {S :: party set. broadcaster ∉ S ∧ card S >= ⌈real n / 2⌉}
  using thresh by (intro exI[of - ?S]) simp
qed

```

```

typedef classic-quorum =
  {S :: party set. broadcaster ∉ S ∧ card S >= ⌈real (n + f - 1) / 2⌉}
proof -

```

Proof sketch: use $UNIV - \{broadcaster\}$ once more. The global inequalities *fault-bound* and $2 \leq n$ imply that $n - 1$ dominates the quorum threshold.

```

let ?S = UNIV - {broadcaster} :: party set
have card-S: card ?S = n - 1
  by (simp add: n-def)
have thresh: card ?S >= ⌈real (n + f - 1) / 2⌉
proof -
  have real (n - 1) >= real (n + f - 1) / 2
    using fault-bound two-parties by linarith
  then have ⌈real (n - 1)⌉ >= ⌈real (n + f - 1) / 2⌉
    by (intro ceiling-mono)
  then have n - 1 >= ⌈real (n + f - 1) / 2⌉
    by simp
  then show ?thesis
    by (simp add: card-S)
qed
show ∃ S :: party set.
  S ∈ {S :: party set. broadcaster ∉ S ∧ card S >= ⌈real (n + f - 1) / 2⌉}
  using thresh by (intro exI[of - ?S]) simp
qed

```

```

typedef amplification-quorum =
  {S :: party set. card S >= f + 1}

```

proof –

Proof sketch: the full universe has cardinality n , and the global inequality *fault-bound* implies $f + 1 \leq n$.

```

have thresh: card (UNIV :: party set) >= f + 1
proof –
  have n >= f + 1
    using fault-bound by linarith
  then show ?thesis
    by (simp add: n-def)
qed
show  $\exists S :: \text{party set}. S \in \{S :: \text{party set}. \text{card } S \geq f + 1\}$ 
  using thresh by (intro exI[of - UNIV :: party set]) simp
qed

```

```

typedef commit-quorum =
  {S :: party set. card S >= 2 * f + 1}
proof –

```

Proof sketch: the full universe again witnesses nonemptiness because the global inequality *fault-bound* implies $2 * f + 1 \leq n$.

```

have thresh: card (UNIV :: party set) >= 2 * f + 1
proof –
  have n >= 2 * f + 1
    using fault-bound by linarith
  then show ?thesis
    by (simp add: n-def)
qed
show  $\exists S :: \text{party set}. S \in \{S :: \text{party set}. \text{card } S \geq 2 * f + 1\}$ 
  using thresh by (intro exI[of - UNIV :: party set]) simp
qed

```

2 Intersection properties

```

lemma card-lemma-1:
  assumes broadcaster ∈ faulty
  and broadcaster ∉ S1
  and card S1 ≥ ⌈real (n + 2*f - 2) / 2⌉
  and broadcaster ∉ S2
  and card S2 ≥ ⌈real n / 2⌉
  shows (S1 ∩ S2) - faulty ≠ {}

```

Proof sketch: Since broadcaster is not in S_1 or S_2 , both are subsets of $UNIV - \{\text{broadcaster}\}$ which has cardinality $n - 1$. By inclusion-exclusion, $\text{card } (S_1 \cap S_2)$ is at least f . Since broadcaster is in *faulty* but not in $S_1 \cap S_2$, at most $f - 1$ faulty elements can be in the intersection. So $S_1 \cap S_2 - \text{faulty}$ has at least 1 element.

proof –

have $n\text{-pos}$: $n \geq 1$ **using** *fault-bound* **by** (*simp add: n-def*)
have $f\text{-pos}$: $f \geq 1$
by (*metis One-nat-def* $\langle \text{broadcaster} \in \text{faulty} \rangle$ *bot-nat-0.extremum-unique card.remove*
f-def finite nat.simps(3) not-less-eq-eq)
have $\text{card-}U$: $\text{card } (UNIV - \{\text{broadcaster}\}) :: \text{party set} = n - 1$
using $n\text{-pos}$ **by** (*simp add: n-def*)

By inclusion-exclusion, the intersection is large enough.

have ie : $\text{card } (S_1 \cap S_2) + (n - 1) \geq \text{card } S_1 + \text{card } S_2$
by (*metis Diff-empty Un-iff add commute add-le-cancel-right* $\langle \text{broadcaster} \notin S_1 \rangle$ $\langle \text{broadcaster} \notin S_2 \rangle$ *card-U card-Un-Int card-mono finite subset-Diff-insert subset-UNIV*)

The sum of cardinalities is at least $n + f - 1$ in int.

have sum-bound : $\text{int } (\text{card } S_1) + \text{int } (\text{card } S_2) \geq \text{int } n + \text{int } f - 1$
proof –
have $n + 2*f \geq 2$ **using** $n\text{-pos}$ $f\text{-pos}$ **by** *linarith*
hence $\text{real } (n + 2*f - 2) / 2 + \text{real } n / 2 = \text{real } n + \text{real } f - 1$
by (*simp add: field-simps*)
thus *?thesis*
using $\langle \text{card } S_1 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil \rangle$ $\langle \text{card } S_2 \geq \lceil \text{real } n / 2 \rceil \rangle$
 $\text{le-of-int-ceiling}[\text{of real } (n + 2*f - 2) / 2]$
 $\text{le-of-int-ceiling}[\text{of real } n / 2]$
by *linarith*
qed

So the intersection has at least f elements.

have inter-card : $\text{int } (\text{card } (S_1 \cap S_2)) \geq \text{int } f$
using ie sum-bound $n\text{-pos}$ **by** *linarith*

Broadcaster is in faulty but not in the intersection, so at most $f - 1$ faulty elements are in the intersection.

have faulty-inter : $\text{card } (\text{faulty} \cap (S_1 \cap S_2)) \leq f - 1$
by (*metis Int-iff Int-lower1 One-nat-def* $\langle \text{broadcaster} \in \text{faulty} \rangle$ $\langle \text{broadcaster} \notin S_1 \rangle$ *bot-nat-0.extremum-unique card.infinite card.remove card-Diff-singleton-if card-seteq f-def f-pos not-less-eq-eq*)

Conclude that the intersection minus faulty is nonempty.

have $\text{card } ((S_1 \cap S_2) - \text{faulty}) \geq 1$
by (*metis Diff-Diff-Int Diff-empty Int-lower2 One-nat-def card.empty card-seteq dec-greater-eq-self-imp-bot empty-subsetI f-def f-pos faulty-inter finite inter-card not-less-eq-eq of-nat-le-iff*)
thus *?thesis* **by** (*metis card.empty not-one-le-zero*)
qed

lemma *card-lemma-2*:

— This follows from *card-lemma-1* because $\lceil \text{real } n / 2 \rceil \leq \lceil \text{real } (n + f - 1) / 2 \rceil$ when $0 < f$.

assumes $\text{broadcaster} \in \text{faulty}$
and $\text{broadcaster} \notin S_1$
and $\text{card } S_1 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil$
and $\text{broadcaster} \notin S_2$
and $\text{card } S_2 \geq \lceil \text{real } (n + f - 1) / 2 \rceil$
shows $(S_1 \cap S_2) - \text{faulty} \neq \{\}$

Since $1 \leq f$, we have $\lceil \text{real } n / 2 \rceil \leq \lceil \text{real } (n + f - 1) / 2 \rceil$, so the assumption on S_2 is at least as strong as what *card-lemma-1* requires. We apply *card-lemma-1* directly.

proof –

have $f \geq 1$
by (*metis One-nat-def* $\langle \text{broadcaster} \in \text{faulty} \rangle$ *bot-nat-0.extremum-unique card.remove* *f-def finite nat.simps(3) not-less-eq-eq*)
hence $\lceil \text{real } (n + f - 1) / 2 \rceil \geq \lceil \text{real } n / 2 \rceil$
by (*intro ceiling-mono divide-right-mono*) *linarith+*
hence $\text{card } S_2 \geq \lceil \text{real } n / 2 \rceil$
using $\langle \text{card } S_2 \geq \lceil \text{real } (n + f - 1) / 2 \rceil \rangle$ **by** *linarith*
thus *?thesis*
using *card-lemma-1*[*OF* $\langle \text{broadcaster} \in \text{faulty} \rangle$ $\langle \text{broadcaster} \notin S_1 \rangle$ $\langle \text{card } S_1 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil \rangle$ $\langle \text{broadcaster} \notin S_2 \rangle$]
by *blast*
qed

lemma *card-lemma-3*:

– This should follow immediately from *card-lemma-1* because $\lceil \text{real } n / 2 \rceil \leq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$ when $0 < f$

assumes $\text{broadcaster} \in \text{faulty}$
and $\text{broadcaster} \notin S_1$
and $\text{card } S_1 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil$
and $\text{broadcaster} \notin S_2$
and $\text{card } S_2 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil$
shows $((S_1 \cap S_2) - \text{faulty}) \neq \{\}$

Since $1 \leq f$, we have $\lceil \text{real } n / 2 \rceil \leq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$, so the assumption on S_2 implies the weaker bound needed by *card-lemma-1*. We apply *card-lemma-1* directly.

proof –

have $f \geq 1$
by (*metis One-nat-def* $\langle \text{broadcaster} \in \text{faulty} \rangle$ *bot-nat-0.extremum-unique card.remove* *f-def finite nat.simps(3) not-less-eq-eq*)
hence $\lceil \text{real } (n + 2*f - 2) / 2 \rceil \geq \lceil \text{real } n / 2 \rceil$
by (*intro ceiling-mono divide-right-mono*) *linarith+*
hence $\text{card } S_2 \geq \lceil \text{real } n / 2 \rceil$
using $\langle \text{card } S_2 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil \rangle$ **by** *linarith*
thus *?thesis*
using *card-lemma-1*[*OF* $\langle \text{broadcaster} \in \text{faulty} \rangle$ $\langle \text{broadcaster} \notin S_1 \rangle$ $\langle \text{card } S_1 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil \rangle$ $\langle \text{broadcaster} \notin S_2 \rangle$]
by *blast*

qed

lemma *card-lemma-4*:

assumes $\text{card } S_1 \geq 2*f + 1$
shows $\text{card } (S_1 - \text{faulty}) \geq f + 1$

At most f elements of S_1 can be faulty, so removing faulty elements leaves at least $2 * f + 1 - f = f + 1$ elements.

proof –

have $\text{card } (S_1 \cap \text{faulty}) \leq f$
by (*metis Int-lower2 card-mono f-def finite*)
moreover have $\text{card } (S_1 - \text{faulty}) = \text{card } S_1 - \text{card } (S_1 \cap \text{faulty})$
by (*metis card-Diff-subset-Int finite*)
ultimately show *?thesis* **using** *assms* **by** *linarith*

qed

lemma *card-lemma-5*:

assumes $\text{broadcaster} \in \text{faulty}$
and $\text{broadcaster} \notin S_1$
and $\text{card } S_1 \geq \lceil \text{real } (n + 2*f - 2) / 2 \rceil$
shows $\text{card } (S_1 - \text{faulty}) \geq \lceil \text{real } n / 2 \rceil$

Since $\text{broadcaster} \in \text{faulty}$ but $\text{broadcaster} \notin S_1$, at most $f - 1$ faulty elements are in S_1 . So $\text{card } (S_1 - \text{faulty})$ is at least $\lceil \text{real } (n + 2 * f - 2) / 2 \rceil - \text{int } (f - 1)$. Since $\lceil \text{real } (n + 2 * f - 2) / 2 \rceil = \lceil \text{real } n / 2 \rceil + \text{int } f - 1$, the result follows.

proof –

have $f\text{-pos}: f \geq 1$
by (*metis One-nat-def <broadcaster ∈ faulty> bot-nat-0.extremum-unique card.remove f-def finite nat.simps(3) not-less-eq-eq*)
have $\text{card } (S_1 \cap \text{faulty}) \leq f - 1$
by (*metis Int-iff Int-lower2 assms(1,2) card.remove card-Diff-singleton-if card-seteq f-def finite not-less-eq-eq*)
moreover have $\text{card } (S_1 - \text{faulty}) = \text{card } S_1 - \text{card } (S_1 \cap \text{faulty})$
by (*metis card-Diff-subset-Int finite*)
moreover have $\lceil \text{real } (n + 2*f - 2) / 2 \rceil = \lceil \text{real } n / 2 \rceil + \text{int } f - 1$

proof –

have $n + 2*f \geq 2$ **using** $f\text{-pos}$ **by** *linarith*
hence $\text{real } (n + 2*f - 2) / 2 = \text{real } n / 2 + \text{real } f - 1$
by (*simp add: field-simps*)
thus *?thesis* **by** *linarith*

qed

moreover have $\text{card } (S_1 \cap \text{faulty}) \leq \text{card } S_1$
by (*simp add: card-mono*)
ultimately have $\text{int } (\text{card } (S_1 - \text{faulty})) \geq \text{int } (\text{card } S_1) - \text{int } (f - 1)$
by (*simp add: card-Diff-subset-Int card-mono*)
moreover have $\text{int } (\text{card } S_1) \geq \lceil \text{real } n / 2 \rceil + \text{int } f - 1$
using *assms(3)* **by** *linarith*
ultimately show *?thesis* **using** $f\text{-pos}$ **by** *linarith*

qed

lemma *card-lemma-6*:

assumes $\text{card } S_1 \geq \lceil \text{real } n / 2 \rceil$
shows $\exists p \in S_1 . p \notin \text{faulty}$

Proof sketch: From $3 * f < n$ we get $\text{int } f < \lceil \text{real } n / 2 \rceil$, so $\text{card } \text{faulty} < \text{card } S_1$. Hence S_1 cannot be a subset of faulty , meaning there exists a member of S_1 not in faulty .

proof –

have $\text{int } f < \lceil \text{real } n / 2 \rceil$

proof –

have $2 * f < n$ **using** *fault-bound* **by** *linarith*

hence $\text{real } (2*f) < \text{real } n$ **by** *simp*

hence $\text{real } f < \text{real } n / 2$ **by** *simp*

thus *?thesis* **by** *linarith*

qed

hence $\text{int } f < \text{int } (\text{card } S_1)$ **using** *assms* **by** *linarith*

hence $\text{card } \text{faulty} < \text{card } S_1$ **by** (*simp add: f-def*)

hence $\neg S_1 \subseteq \text{faulty}$

by (*metis card-mono finite leD*)

thus *?thesis* **by** *blast*

qed

lemma *card-lemma-7*:

assumes $\text{broadcaster} \in \text{faulty}$

and $\text{broadcaster} \notin S_1$

and $\text{card } S_1 \geq \lceil \text{real } (n + f - 1) / 2 \rceil$

and $\text{broadcaster} \notin S_2$

and $\text{card } S_2 \geq \lceil \text{real } (n + f - 1) / 2 \rceil$

shows $(S_1 \cap S_2) - \text{faulty} \neq \{\}$

Proof sketch: Since *broadcaster* is excluded from S_1 and S_2 , both sets lie in $\text{UNIV} - \{\text{broadcaster}\}$, whose cardinality is $n - 1$. Inclusion-exclusion and the lower bounds on $\text{card } S_1$ and $\text{card } S_2$ yield $f \leq \text{card } (S_1 \cap S_2)$. Because $\text{broadcaster} \in \text{faulty}$ but $\text{broadcaster} \notin S_1 \cap S_2$, at most $f - 1$ faulty parties lie in the intersection. Therefore $S_1 \cap S_2 - \text{faulty}$ is nonempty.

proof –

have $n\text{-pos}: n \geq 1$

using *fault-bound* **by** (*simp add: n-def*)

have $f\text{-pos}: f \geq 1$

by (*metis One-nat-def* $\langle \text{broadcaster} \in \text{faulty} \rangle$ *bot-nat-0.extremum-unique* *card.remove* *f-def* *finite* *nat.simps(3)* *not-less-eq-eq*)

have $\text{card-}U: \text{card } (\text{UNIV} - \{\text{broadcaster}\}) :: \text{party set} = n - 1$

using $n\text{-pos}$ **by** (*simp add: n-def*)

have $\text{union-bound}: \text{card } (S_1 \cup S_2) \leq n - 1$

by (*metis Diff-empty Un-iff* *assms(2,4)* *card-U* *card-mono* *finite* *subset-Diff-insert* *top-greatest*)

```

have ie: card (S1 ∩ S2) + (n - 1) ≥ card S1 + card S2
  by (metis add.commute add-le-cancel-right card-Un-Int finite union-bound)
have sum-bound: int (card S1) + int (card S2) ≥ int n + int f - 1
proof -
  have 2 * ⌈real (n + f - 1) / 2⌉ ≥ int n + int f - 1
    using le-of-int-ceiling[of real (n + f - 1) / 2] by linarith
  thus ?thesis
    using ⟨card S1 ≥ ⌈real (n + f - 1) / 2⌉⟩ ⟨card S2 ≥ ⌈real (n + f - 1) / 2⌉⟩
    by linarith
qed
have inter-card: card (S1 ∩ S2) ≥ f
  using ie sum-bound n-pos by linarith
have faulty-inter: card (faulty ∩ (S1 ∩ S2)) ≤ f - 1
  by (metis Int-iff Int-lower1 One-nat-def assms(1,2) card.infinite card.remove
    card-Diff-singleton-if card-seteq dual-order.refl f-def f-pos not-less-eq-eq)
show ?thesis
  by (metis Diff-eq-empty-iff Int-absorb1 diff-diff-cancel diff-is-0-eq diff-le-mono2
    f-pos faulty-inter inter-card not-one-le-zero)
qed

```

3 Proof that we have an instance of the abstract domain model

definition *opt-quorum-member* :: *opt-quorum* => *party* => *bool* **where**
opt-quorum-member Q p ≡ p ∈ Rep-opt-quorum Q

definition *maj-quorum-member* :: *maj-quorum* => *party* => *bool* **where**
maj-quorum-member Q p ≡ p ∈ Rep-maj-quorum Q

definition *classic-quorum-member* :: *classic-quorum* => *party* => *bool* **where**
classic-quorum-member Q p ≡ p ∈ Rep-classic-quorum Q

definition *amplification-quorum-member* :: *amplification-quorum* => *party* => *bool* **where**
amplification-quorum-member Q p ≡ p ∈ Rep-amplification-quorum Q

definition *commit-quorum-member* :: *commit-quorum* => *party* => *bool* **where**
commit-quorum-member Q p ≡ p ∈ Rep-commit-quorum Q

interpretation *axiomatic-abstract-domain-model*:

abstract-domain-model broadcaster faulty *opt-quorum-member* *maj-quorum-member*
classic-quorum-member *amplification-quorum-member* *commit-quorum-member*

proof (standard)

show ¬ *classic-quorum-member* quQ broadcaster **for** quQ
using Rep-classic-quorum[of quQ] **unfolding** *classic-quorum-member-def* **by**
auto

show $\neg \text{maj-quorum-member } vtQ \text{ broadcaster for } vtQ$
using *Rep-maj-quorum*[of *vtQ*] **unfolding** *maj-quorum-member-def* **by** *auto*

show $\neg \text{opt-quorum-member } opQ \text{ broadcaster for } opQ$
using *Rep-opt-quorum*[of *opQ*] **unfolding** *opt-quorum-member-def* **by** *auto*

show $\exists coQ. \forall p. \text{commit-quorum-member } coQ \ p \longrightarrow p \notin \text{faulty}$

Proof sketch: *UNIV - faulty* has cardinality $n - f$, and *fault-bound* implies $2 * f + 1 \leq n - f$. Using this set as the carrier of a commit quorum yields a witness whose members are all nonfaulty.

proof –
let *?S* = (*UNIV :: party set*) – *faulty*
have *card-S*: *card ?S* = $n - f$
by (*simp add: n-def f-def Finite-Set.card-Diff-subset*)
have *co-carrier*: *?S* $\in \{S :: \text{party set}. \text{card } S \geq 2 * f + 1\}$
proof
have $n - f \geq 2 * f + 1$
using *fault-bound* **by** *linarith*
with *card-S* **show** *card ?S* $\geq 2 * f + 1$
by *simp*
qed
let *?coQ* = *Abs-commit-quorum ?S*
have *rep-coQ*: *Rep-commit-quorum ?coQ* = *?S*
using *co-carrier* **by** (*simp add: Abs-commit-quorum-inverse*)
show *?thesis*
using *rep-coQ* **unfolding** *commit-quorum-member-def* **by** *auto*
qed

show $\exists amQ. \forall p. \text{amplification-quorum-member } amQ \ p \longrightarrow p \notin \text{faulty}$

Proof sketch: *UNIV - faulty* also has enough members for the amplification threshold because *fault-bound* implies $f + 1 \leq n - f$. Turning that set into an amplification quorum gives the desired witness.

proof –
let *?S* = (*UNIV :: party set*) – *faulty*
have *card-S*: *card ?S* = $n - f$
by (*simp add: n-def f-def Finite-Set.card-Diff-subset*)
have *am-carrier*: *?S* $\in \{S :: \text{party set}. \text{card } S \geq f + 1\}$
proof
have $n - f \geq f + 1$
using *fault-bound* **by** *linarith*
with *card-S* **show** *card ?S* $\geq f + 1$
by *simp*
qed
let *?amQ* = *Abs-amplification-quorum ?S*
have *rep-amQ*: *Rep-amplification-quorum ?amQ* = *?S*
using *am-carrier* **by** (*simp add: Abs-amplification-quorum-inverse*)
show *?thesis*

using *rep-amQ* **unfolding** *amplification-quorum-member-def* **by** *auto*
qed

show $\exists vtQ. \forall p. \text{maj-quorum-member } vtQ \ p \longrightarrow p \notin \text{faulty}$

Proof sketch: Use the nonfaulty non-broadcaster parties as the carrier. If the broadcaster is faulty this is $UNIV - \text{faulty}$; otherwise it is $UNIV - \text{faulty} - \{\text{broadcaster}\}$. In both cases the resulting set still reaches the majority threshold.

proof –

let $?S = ((UNIV :: \text{party set}) - \text{faulty}) - \{\text{broadcaster}\}$

have *card-nonfaulty*: $\text{card } ((UNIV :: \text{party set}) - \text{faulty}) = n - f$

by (*simp add: n-def f-def Finite-Set.card-Diff-subset*)

have *vt-carrier*: $?S \in \{S :: \text{party set}. \text{broadcaster} \notin S \wedge \text{card } S \geq \lceil \text{real } n / 2 \rceil\}$

proof –

have *broadcaster-notin*: $\text{broadcaster} \notin ?S$

by *simp*

have *card-bound*: $\text{card } ?S \geq \lceil \text{real } n / 2 \rceil$

proof (*cases broadcaster $\in$ faulty*)

case *True*

have *card-S*: $\text{card } ?S = n - f$

using *True card-nonfaulty* **by** *simp*

have *real (n - f) $\geq$ real n / 2*

using *fault-bound* **by** *linarith*

then have $\lceil \text{real } (n - f) \rceil \geq \lceil \text{real } n / 2 \rceil$

by (*intro ceiling-mono*)

with *card-S* **show** *?thesis*

by *simp*

next

case *False*

have *card-S*: $\text{card } ?S = n - f - 1$

using *False card-nonfaulty* **by** (*simp add: Finite-Set.card-Diff-singleton*)

have *n-bound*: $2 * f + 2 \leq n$

proof (*cases f = 0*)

case *True*

then show *?thesis* **using** *two-parties* **by** *simp*

next

case *False*

then have $f \geq 1$ **by** *simp*

then show *?thesis* **using** *fault-bound* **by** *linarith*

qed

have *real (n - f - 1) $\geq$ real n / 2*

using *n-bound* **by** *linarith*

then have $\lceil \text{real } (n - f - 1) \rceil \geq \lceil \text{real } n / 2 \rceil$

by (*intro ceiling-mono*)

with *card-S* **show** *?thesis*

by *simp*

qed

show *?thesis*

```

    using broadcaster-notin card-bound by simp
  qed
  let ?vtQ = Abs-maj-quorum ?S
  have rep-vtQ: Rep-maj-quorum ?vtQ = ?S
    using vt-carrier by (simp add: Abs-maj-quorum-inverse)
  show ?thesis
    using rep-vtQ unfolding maj-quorum-member-def by auto
  qed

```

show $\exists quQ. \forall p. \text{classic-quorum-member } quQ\ p \longrightarrow p \notin \text{faulty}$

Proof sketch: Reuse the nonfaulty non-broadcaster parties as the carrier. The same case split on whether *broadcaster* is faulty shows that this set also meets the classic-quorum threshold, so it yields a classic quorum all of whose members are nonfaulty.

```

proof -
  let ?S = ((UNIV :: party set) - faulty) - {broadcaster}
  have card-nonfaulty: card ((UNIV :: party set) - faulty) = n - f
    by (simp add: n-def f-def Finite-Set.card-Diff-subset)
  have qu-carrier: ?S ∈ {S :: party set. broadcaster ∉ S ∧ card S ≥ ⌈real (n + f - 1) / 2⌉}
  proof -
    have broadcaster-notin: broadcaster ∉ ?S
      by simp
    have card-bound: card ?S ≥ ⌈real (n + f - 1) / 2⌉
    proof (cases broadcaster ∈ faulty)
      case True
        have card-S: card ?S = n - f
          using True card-nonfaulty by simp
        have real (n - f) ≥ real (n + f - 1) / 2
          using fault-bound by linarith
        then have ⌈real (n - f)⌉ ≥ ⌈real (n + f - 1) / 2⌉
          by (intro ceiling-mono)
        with card-S show ?thesis
          by simp
      case False
        have card-S: card ?S = n - f - 1
          using False card-nonfaulty by (simp add: Finite-Set.card-Diff-singleton)
        have n-bound: 3 * f + 1 ≤ n
          using fault-bound by linarith
        have real (n - f - 1) ≥ real (n + f - 1) / 2
          using n-bound by linarith
        then have ⌈real (n - f - 1)⌉ ≥ ⌈real (n + f - 1) / 2⌉
          by (intro ceiling-mono)
        with card-S show ?thesis
          by simp
    qed
  next
    case False
    have card-S: card ?S = n - f - 1
      using False card-nonfaulty by (simp add: Finite-Set.card-Diff-singleton)
    have n-bound: 3 * f + 1 ≤ n
      using fault-bound by linarith
    have real (n - f - 1) ≥ real (n + f - 1) / 2
      using n-bound by linarith
    then have ⌈real (n - f - 1)⌉ ≥ ⌈real (n + f - 1) / 2⌉
      by (intro ceiling-mono)
    with card-S show ?thesis
      by simp
  qed

```

```

  show ?thesis
    using broadcaster-notin card-bound by simp
qed
let ?quQ = Abs-classic-quorum ?S
have rep-quQ: Rep-classic-quorum ?quQ = ?S
  using qu-carrier by (simp add: Abs-classic-quorum-inverse)
show ?thesis
  using rep-quQ unfolding classic-quorum-member-def by auto
qed

```

show $\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } \text{opQ } p$ for opQ

Proof sketch: $\text{Rep-opt-quorum } \text{opQ}$ is too large to fit inside faulty . The optimistic threshold forces $\text{int } f < \max 1 \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$, and the representing set has at least that many members, so it must contain a nonfaulty party.

```

proof -
  have opQ-card:  $\max 1 \lceil \text{real } (n + 2 * f - 2) / 2 \rceil \leq \text{int } (\text{card } (\text{Rep-opt-quorum } \text{opQ}))$ 
  using Rep-opt-quorum[of opQ] by auto
  have opt-thresh:  $\text{int } f < \max 1 \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$ 
  proof (cases  $f = 0$ )
    case True
      then show ?thesis by simp
    next
      case False
        then have f-pos:  $f \geq 1$  by simp
        have  $4 \leq n$ 
          using fault-bound f-pos by linarith
        then have  $\text{real } f < \text{real } (n + 2 * f - 2) / 2$ 
          by linarith
        then have  $\text{int } f < \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$ 
          by (simp add: less-ceiling-iff)
        then show ?thesis
          by simp
      qed
  have  $\text{int } (\text{card } \text{faulty}) < \text{int } (\text{card } (\text{Rep-opt-quorum } \text{opQ}))$ 
  proof -
    have  $\text{int } f < \text{int } (\text{card } (\text{Rep-opt-quorum } \text{opQ}))$ 
      using opQ-card opt-thresh by linarith
    then show ?thesis
      by (simp add: f-def)
  qed
  then have  $\text{card } \text{faulty} < \text{card } (\text{Rep-opt-quorum } \text{opQ})$ 
    by simp
  then have  $\neg \text{Rep-opt-quorum } \text{opQ} \subseteq \text{faulty}$ 
    by (metis card-mono finite leD)
  then obtain  $p$  where  $p \in \text{Rep-opt-quorum } \text{opQ}$  and  $p \notin \text{faulty}$ 
    by blast

```

then show *?thesis*
unfolding *opt-quorum-member-def* **by** *blast*
qed

show $\text{broadcaster} \in \text{faulty} \longrightarrow (\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } \text{opQ}_1 \ p \wedge \text{opt-quorum-member } \text{opQ}_2 \ p)$ **for** *opQ₁ opQ₂*

Proof sketch: Apply *card-lemma-3* to the representing sets *Rep-opt-quorum opQ₁* and *Rep-opt-quorum opQ₂*. Any witness in their nonfaulty intersection immediately yields the required quorum members.

proof
assume *broadcaster-faulty: broadcaster ∈ faulty*
have *opQ1-props*:
 $\text{broadcaster} \notin \text{Rep-opt-quorum } \text{opQ}_1$
 $\text{card } (\text{Rep-opt-quorum } \text{opQ}_1) \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
using *Rep-opt-quorum[of opQ₁]* **by** *auto*
have *opQ2-props*:
 $\text{broadcaster} \notin \text{Rep-opt-quorum } \text{opQ}_2$
 $\text{card } (\text{Rep-opt-quorum } \text{opQ}_2) \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
using *Rep-opt-quorum[of opQ₂]* **by** *auto*
from *card-lemma-3[OF broadcaster-faulty opQ1-props(1) opQ1-props(2) opQ2-props(1) opQ2-props(2)]*
obtain *p* **where** $p \in (\text{Rep-opt-quorum } \text{opQ}_1 \cap \text{Rep-opt-quorum } \text{opQ}_2) - \text{faulty}$
by *blast*
then show $\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } \text{opQ}_1 \ p \wedge \text{opt-quorum-member } \text{opQ}_2 \ p$
unfolding *opt-quorum-member-def* **by** *blast*
qed

show $\text{broadcaster} \in \text{faulty} \longrightarrow (\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } \text{opQ} \ p \wedge \text{maj-quorum-member } \text{vtQ} \ p)$ **for** *opQ vtQ*

Proof sketch: Apply *card-lemma-1* to *Rep-opt-quorum opQ* and *Rep-maj-quorum vtQ*. A witness in the nonfaulty intersection satisfies both membership predicates after unfolding their definitions.

proof
assume *broadcaster-faulty: broadcaster ∈ faulty*
have *opQ-props*:
 $\text{broadcaster} \notin \text{Rep-opt-quorum } \text{opQ}$
 $\text{card } (\text{Rep-opt-quorum } \text{opQ}) \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
using *Rep-opt-quorum[of opQ]* **by** *auto*
have *vtQ-props*:
 $\text{broadcaster} \notin \text{Rep-maj-quorum } \text{vtQ}$
 $\text{card } (\text{Rep-maj-quorum } \text{vtQ}) \geq \lceil \text{real } n / 2 \rceil$
using *Rep-maj-quorum[of vtQ]* **by** *auto*
from *card-lemma-1[OF broadcaster-faulty opQ-props(1) opQ-props(2) vtQ-props(1) vtQ-props(2)]*
obtain *p* **where** $p \in (\text{Rep-opt-quorum } \text{opQ} \cap \text{Rep-maj-quorum } \text{vtQ}) - \text{faulty}$
by *blast*

then show $\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } opQ\ p \wedge \text{maj-quorum-member } vtQ\ p$
unfolding *opt-quorum-member-def* *maj-quorum-member-def* **by** *blast*
qed

show $\text{broadcaster} \in \text{faulty} \longrightarrow (\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } opQ\ p \wedge \text{classic-quorum-member } quQ\ p)$ **for** $opQ\ quQ$

Proof sketch: Use *card-lemma-2* on *Rep-opt-quorum* opQ and *Rep-classic-quorum* quQ . The produced witness lies in both representing sets and outside *faulty*.

proof
assume *broadcaster-faulty*: $\text{broadcaster} \in \text{faulty}$
have *opQ*-props:
 $\text{broadcaster} \notin \text{Rep-opt-quorum } opQ$
 $\text{card } (\text{Rep-opt-quorum } opQ) \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$
using *Rep-opt-quorum*[*of opQ*] **by** *auto*
have *quQ*-props:
 $\text{broadcaster} \notin \text{Rep-classic-quorum } quQ$
 $\text{card } (\text{Rep-classic-quorum } quQ) \geq \lceil \text{real } (n + f - 1) / 2 \rceil$
using *Rep-classic-quorum*[*of quQ*] **by** *auto*
from *card-lemma-2*[*OF broadcaster-faulty opQ*-props(1) *opQ*-props(2) *quQ*-props(1) *quQ*-props(2)]
obtain $p \text{ where } p \in (\text{Rep-opt-quorum } opQ \cap \text{Rep-classic-quorum } quQ) - \text{faulty}$
by *blast*
then show $\exists p. p \notin \text{faulty} \wedge \text{opt-quorum-member } opQ\ p \wedge \text{classic-quorum-member } quQ\ p$
unfolding *opt-quorum-member-def* *classic-quorum-member-def* **by** *blast*
qed

show $\text{broadcaster} \in \text{faulty} \longrightarrow (\exists p. p \notin \text{faulty} \wedge \text{classic-quorum-member } quQ_1\ p \wedge \text{classic-quorum-member } quQ_2\ p)$ **for** $quQ_1\ quQ_2$

Proof sketch: Use *card-lemma-7* on *Rep-classic-quorum* quQ_1 and *Rep-classic-quorum* quQ_2 . The lemma yields a party in both representing sets and outside *faulty*, which is exactly the desired witness.

proof
assume *broadcaster-faulty*: $\text{broadcaster} \in \text{faulty}$
from *card-lemma-7*[*OF broadcaster-faulty*]
obtain $p \text{ where } p \in (\text{Rep-classic-quorum } quQ_1 \cap \text{Rep-classic-quorum } quQ_2) - \text{faulty}$
using *Rep-classic-quorum*[*of quQ₁*] *Rep-classic-quorum*[*of quQ₂*] **by** *auto*
then show $\exists p. p \notin \text{faulty} \wedge \text{classic-quorum-member } quQ_1\ p \wedge \text{classic-quorum-member } quQ_2\ p$
unfolding *classic-quorum-member-def* **by** *blast*
qed

show $\exists p. p \notin \text{faulty} \wedge \text{classic-quorum-member } Q\ p$ **for** Q

Proof sketch: Every classic quorum is too large to be contained in *faulty*. Its

threshold implies $\text{int } f < \lceil \text{real } (n + f - 1) / 2 \rceil$, while the representing set of Q has at least that many members, so it contains a nonfaulty party.

```

proof –
  have  $Q\text{-card}$ :  $\lceil \text{real } (n + f - 1) / 2 \rceil \leq \text{int } (\text{card } (\text{Rep-classic-quorum } Q))$ 
    using  $\text{Rep-classic-quorum}$ [of  $Q$ ] by auto
  have  $qu\text{-thresh}$ :  $\text{int } f < \lceil \text{real } (n + f - 1) / 2 \rceil$ 
  proof (cases  $f = 0$ )
    case True
      have  $0 < \text{real } (n + f - 1) / 2$ 
        using two-parties True by linarith
      then show ?thesis
        using True by simp
    next
      case False
      then have  $f\text{-pos}$ :  $f \geq 1$  by simp
      have  $f + 1 < n$ 
        using fault-bound f-pos by linarith
      then have  $\text{real } f < \text{real } (n + f - 1) / 2$ 
        by linarith
      then show ?thesis
        by (simp add: less-ceiling-iff)
    qed
  have  $\text{card } \text{faulty} < \text{card } (\text{Rep-classic-quorum } Q)$ 
    using  $Q\text{-card } qu\text{-thresh}$  by (simp add: f-def)
  then have  $\neg \text{Rep-classic-quorum } Q \subseteq \text{faulty}$ 
    by (metis card-mono finite leD)
  then obtain  $p$  where  $p \in \text{Rep-classic-quorum } Q$  and  $p \notin \text{faulty}$ 
    by blast
  then show ?thesis
    unfolding classic-quorum-member-def by blast
  qed

```

show $\exists p. p \notin \text{faulty} \wedge \text{maj-quorum-member } vtQ\ p$ **for** vtQ

Proof sketch: *card-lemma-6* applies directly to *Rep-maj-quorum* vtQ , whose cardinality is part of the typedef invariant.

```

using Rep-maj-quorum card-lemma-6 maj-quorum-member-def by fastforce

```

show $\exists p. p \notin \text{faulty} \wedge \text{amplification-quorum-member } amQ\ p$ **for** amQ

Proof sketch: *Rep-amplification-quorum* amQ has at least $f + 1$ elements, while *faulty* has exactly f . Therefore the representing set cannot be contained in *faulty*.

```

proof –
  have  $amQ\text{-props}$ :  $\text{card } (\text{Rep-amplification-quorum } amQ) \geq f + 1$ 
    using Rep-amplification-quorum[of  $amQ$ ] by auto
  have  $\text{card } \text{faulty} < \text{card } (\text{Rep-amplification-quorum } amQ)$ 
    using  $amQ\text{-props}$  by (simp add: f-def)

```

```

then have  $\neg \text{Rep-amplification-quorum } amQ \subseteq \text{faulty}$ 
  by (metis card-mono finite leD)
then obtain  $p$  where  $p \in \text{Rep-amplification-quorum } amQ$  and  $p \notin \text{faulty}$ 
  by blast
then show ?thesis
  unfolding amplification-quorum-member-def by blast
qed

```

show $\text{broadcaster} \in \text{faulty} \longrightarrow (\exists vtQ. \forall p. \text{maj-quorum-member } vtQ\ p \longrightarrow p \notin \text{faulty} \wedge \text{opt-quorum-member } opQ\ p)$ **for** opQ

Proof sketch: *card-lemma-5* shows that $\text{Rep-opt-quorum } opQ - \text{faulty}$ still meets the vote threshold. Build a vote quorum from that set, so every one of its members is both nonfaulty and already in the optimistic quorum.

```

proof
  assume broadcaster-faulty:  $\text{broadcaster} \in \text{faulty}$ 
  let  $?S = \text{Rep-opt-quorum } opQ - \text{faulty}$ 
  have opQ-props:
     $\text{broadcaster} \notin \text{Rep-opt-quorum } opQ$ 
     $\text{card } (\text{Rep-opt-quorum } opQ) \geq \lceil \text{real } (n + 2 * f - 2) / 2 \rceil$ 
    using Rep-opt-quorum[of opQ] by auto
  have vote-carrier:  $?S \in \{S :: \text{party set. } \text{broadcaster} \notin S \wedge \text{card } S \geq \lceil \text{real } n / 2 \rceil\}$ 
  using opQ-props card-lemma-5[OF broadcaster-faulty opQ-props(1) opQ-props(2)]
  by auto
  let  $?vtQ = \text{Abs-maj-quorum } ?S$ 
  have rep-vtQ:  $\text{Rep-maj-quorum } ?vtQ = ?S$ 
  using vote-carrier by (simp add: Abs-maj-quorum-inverse)
  show  $\exists vtQ. \forall p. \text{maj-quorum-member } vtQ\ p \longrightarrow p \notin \text{faulty} \wedge \text{opt-quorum-member } opQ\ p$ 
  using opt-quorum-member-def rep-vtQ maj-quorum-member-def by auto
qed

```

show $\exists amQ. \forall p. \text{amplification-quorum-member } amQ\ p \longrightarrow p \notin \text{faulty} \wedge \text{commit-quorum-member } coQ\ p$ **for** coQ

Proof sketch: *card-lemma-4* guarantees that $\text{Rep-commit-quorum } coQ - \text{faulty}$ still has the amplification threshold. Use that set as the representing set of a witness amplification quorum.

```

proof –
  let  $?S = \text{Rep-commit-quorum } coQ - \text{faulty}$ 
  have coQ-props:  $\text{card } (\text{Rep-commit-quorum } coQ) \geq 2 * f + 1$ 
  using Rep-commit-quorum[of coQ] by auto
  have am-carrier:  $?S \in \{S :: \text{party set. } \text{card } S \geq f + 1\}$ 
  using card-lemma-4[OF coQ-props] by simp
  let  $?amQ = \text{Abs-amplification-quorum } ?S$ 
  have rep-amQ:  $\text{Rep-amplification-quorum } ?amQ = ?S$ 
  using am-carrier by (simp add: Abs-amplification-quorum-inverse)
  show ?thesis

```

```
      using amplification-quorum-member-def commit-quorum-member-def rep-amQ
by auto
  qed
qed
end
```